Protocol specification

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OPEN PROTOCOL SPECIFICATION

Version 2.55

Info: http://www.duotecno.be

Introduction:

Different messages:

Three types of messages: commands, responses and notifications.

The client sends commands which have two types:
- A control message which requests the server to perform an action.
- A query message to request information from the server.

The server sends:
- An ACK/NAK control message which indicates the server received the messages and is ready to receive a new message. (only for RS232 protocol.)
- A response message that contains information requested by the query message.
- A notification message which is send spontaneous and indicates an event occurred.

These messages can be grouped by functionality:
- Connection management and status messages.
- Node database management and status messages.
- Home automation control, status and request messages.

All messages are packetized in an ASCII command string message. This command string message provides synchronization and identification. When the message holds characters not conform to the packet structure the server closes the connection.

Multiple command strings can be send via one message. The server processes the received data when a end of message character (=Line Feed) is received. **The maximum message length is restricted to 250 bytes.**

Packet structure:

All data is formatted in a ASCII command string format. A commando string starts via a start character and ends with an end character. All data values are separated via a separator character.

Start command string character.

Fixed character '[' used for synchronization.

Data values:

The data values are represented as decimal byte values (0 - 255). Larger integers which are represented by multiple bytes are always transmitted and interpreted as most significant byte first, followed by the least significant byte.

Separator character:

Fixed character ',' used to separate the data values.

End command string character:

Fixed character ‘]’ used to signal the command string has ended.

End of message character

Fixed character ‘LF’ (h0a) used to signal the server can process the received command strings.
For compatibility reasons ‘CR’ + ‘LF’ pair is allowed. (h0d,h0a).

Example:

Command string = [163,2,3,0]

Byte#	Char	Description
0	‘[‘	Start char
1	‘1’	Data: Command
2	‘6’	Data: Command
3	‘3’	Data: Command
4	‘,’	Separator
5	‘2’	Data: Method
6	‘,’	Separator
7	‘3’	Node Address
8	‘,’	Separator
9	‘0’	Unit address
10	‘]’	End command string
11	LF (h0A)	End character.

Supported communication media:

RS232 protocol

There are additional ACK/NAK handshaking commands required for RS232 communication. When the server receives a good command it will reply with a ACK message [0,1]. When the server receives a message that it cannot parse it will reply with a NAK message [0,0]. The client must always wait for a response from the server. When the client receives no answer within 10 seconds it must retry sending the command.

TCP/IP Protocol

The TCP/IP communication is password protected. The client must send a login request and wait for a response from the server. As long as authentication is pending, the server will not respond to any other commands. The login request command is the first command that must be send.

Commands overview.

Attention:
Verify the protocol version before sending commands to the home server.
Since protocol v2.04 the protocol and software versions can be requested via the protocol. For all versions this info can be checked by the Duotecno configurator tool.

Since protocol version v2.04 the home server node can be set the master in the home automation network. This version supports commands for home automation network management and configuration.
When the homeserver node is the master it will also send events when an IR RX unit receives IR codes.

Protocol version v2.04 also supports commands to set and request time/date and week scheduling.

Protocol version v2.05 added new commands and status messages specific for 3rd party control of temperature control system. The new methods and status messages have been added to the existing sensor unit.

Since protocol version v2.06 there have been added nodes that support a LITE version of this protocol.
The nodes that support the LITE version have software version v90.XX.
The nodes that support the FULL version have software version v65.XX.

The nodes that support the full version of protocol version v2.06 have support for bidirectional control of A/V matrix systems.

In protocol version v2.07 there are more possibilities to integrate virtual units. These units can be used to control 3rd party systems.

Id	Name	Since protocol:	Changes protocol:	Note
4	EV_UNITCONTROLSTATUS	V2.02	--	
5	EV_UNITDIMSTATUS	V2.02	--	
6	EV_UNITSWITCHSTATUS	V2.02	--	
7	EV_UNITSENSSTATUS	V2.02	V2.05	*4*
17	EV_MESSAGEERROR	V2.03	--	
18	EV_NODERESET	V2.04	--	*1*
23	EV_UNITAUDIOSTATUS	V2.02	--	
38	EV_UNITDUOSWITCHSTATUS	V2.02	--	
54	EV_UNITAVMATRIXSTATUS	**V2.06**	--	
48	EV_UNITDEFAULTSTATUS	V2.02	--	
64	EV_NODEDATABASEINFO	V2.02	--	
66	EV_APPLICATIONTASKSTATUS	V2.04	--	*1*
67	EV_CLIENTCONNECTSET	V2.02	--	
69	EV_UNITMACROCOMMAND	V2.02	**V2.06**	*1*
70	EV_UNITAUDIOEXTSTATUS	V2.02	--	
70	EV_HEARTBEATSTATUS_OLD	V2.03	--	*2*
71	EV_TIMEDATESTATUS	V2.04	--	*1*
72	EV_HEARTBEATSTATUS	V2.04	--	

73	EV_SCHEDULESTATUS	V2.04	--	
74	EV_NODEMANAGEMENTINFO	V2.04	--	
75	EV_ACCESSLEVELSET	V2.04	--	*1*
76	EV_VIDEOPHONESTATUS	V2.04	--	*1*
77	EV_REGISTERMAP	V2.09		*1*
131	FC_UNITDIMREQUESTSTATUS	V2.02	--	
136	FC_UNITSENSSET	V2.02	V2.05	
137	FC_UNITREQUESTSENSSTATUS	V2.02	V2.05	
155	FC_NODERESETSET	V2.04	--	*1*
159	FC_UNITAUDIOBASICSET	V2.02	--	
162	FC_UNITDIMSET	V2.02	--	
163	FC_UNITSWITCHSET	V2.02	--	
168	FC_UNITCONTROLSET	V2.02	--	
170	FC_TIMEDATE	V2.04	--	
173	FC_UNITIRTXSET	V2.02	--	
182	FC_UNITDUOSWITCHSET	V2.02	--	
192	FC_CHECKIRRXCODE	V2.04	--	*1*
193	FC_UNITVIDEOMUXSET	V2.02	--	
202	FC_UNITAVMATRIXSET	V2.02	--	
204	FC_UNITALARMSET	--	--	*3*
208	FC_UNITAUDIOEXTSET	V2.02	--	
209	FC_NODEDATABASEREQUESTSTATUS	V2.02	--	
212	FC_APPLICATIONTASKSET	V2.04	--	*1*
213	FC_REQUESTAPPLICATIONTASKSTATUS	V2.04	--	*1*
214	FC_CLIENTCONNECTSET	V2.02	--	
215	FC_HEARTBEATREQUESTSTATUS	V2.03	--	*2*
216	FC_REQUESTTIMEDATE	V2.04	--	
217	FC_SCHEDULESET	V2.04	--	
218	FC_REQUESTSCHEDULE	V2.04	--	
219	FC_NODEMANAGEMENTSET	V2.04	--	*1*
220	FC_REQUESTNODEMANAGEMENT	V2.04	--	
221	FC_NODEDATABASESET	V2.04	--	*1*
222	FC_ACCESSLEVELSET	V2.04	--	*1*
223	FC_VIDEOPHONESET	V2.04	--	*1*
224	FC_REGISTERMAP	V2.09	--	*1*

Note *1*: These commands are only implemented in the nodes that support the FULL protocol version. The nodes that only support the LITE protocol version will normally reply with EV_MESSAGEERROR + NOT_IMPLEMENTED message.
Note *2*: This commands are deprecated.
Note *3*: This command is reserved but not implemented.
Note *4*: In protocol version 2.07 this command can be used by a 3rd party temp sensor to send his measured temperature value.

Connection management and status frames

Packet fields:

[Command , method , data , data N-1]

Messages:

Client login command and status

The TCP/IP communication is password protected. The client must send a login request and wait for a response from the server. As long as authentication is pending, the server will not respond to any other commands. When the login request holds a wrong password the server will automatically close the connection.

FC_CLIENTCONNECTSET [214]

General:

The client must use this command to send a login request. This is the first command that the client must send to the server.

Message structure:

Frame	Value	Description
Command	214	Command FC_CLIENTCONNECTSET
Method	3	Method: OBJECT_PASSWORD
Data 0	Length	Nr of characters in password data.
Data 1 … N	Password	Password data, max 16 characters

Example:

Example password 'DUOTECNO'.
[214,3,8,68,85,79,84,69,67,78,79]

BYTE#	Char	Description
0	'['	Start char
1	'2'	Data: Command
2	'1'	Data: Command
3	'4'	Data: Command
4	','	Separator
5	'3'	Data: Method
6	','	Separator
7	'8'	Length password
8	','	Separator
9	'6'	D
10	'8'	D
11	','	Separator
12	'8'	U
13	'5'	U

14	‘,’	Separator
15	‘7’	O
16	‘9’	O
17	‘,’	Separator
18	‘8’	T
19	‘4’	T
20	‘,’	Separator
21	‘6’	E
22	‘9’	E
23	‘,’	Separator
24	‘6’	C
25	‘7’	C
26	‘,’	Separator
27	‘7’	N
28	‘8’	N
29	‘,’	Separator
30	‘7’	O
31	‘9’	O
32	‘]’	End command string
33	LF (h0A)	End character.

EV_CLIENTCONNECTSTATUS [67]

General:

The server will reply with this status message to a client login request. When the server sends an access denied command the server will with small delay close the connection. When the server sends the access allowed command normal operation is possible.

Message structure:

Frame	Value	Description
Command	67	Command: EV_CLIENTCONNECTSTATUS.
Method	3	Method: OBJECT_PASSWORD
Data 0	Status	See table password validation.

Table password validation:

Status	Value
ACCESS_DENIED	0
ACCESS_ALLOWED	1

Examples:

Example: Client Allowed
[67,3,1]

BYTE#	Char	Description
0	'['	Start char
1	'6'	Data: Command
2	'7'	Data: Command
4	','	Separator
5	'2'	Data: Method
6	','	Separator
7	'1'	Access allowed
8	']'	End command string
9	LF (h0A)	End character.

Client logout command.

The client can send a log off command. The server will not respond to this connection but will automatically close the connection. This is the proper way to shutdown the connection.

FC_CLIENTCONNECTSET [214]

General:

The client must use this command to disconnect from the server.

Message structure:

Frame	Value	Description
Command	214	Command FC_CLIENTCONNECTSET
Method	0	Method: OBJECT_DISCONNECT

Example:

[214,0]

Heartbeat message.

TCP/IP communication can use a heart beat polling mechanism. When the client sends the request message a reply from the server must be received. When the reply is not received the connection is possible disconnected.

FC_HEARTBEATREQUESTSTATUS [215]

General:

A client uses this message to check the connectivity of the connection. (optional).

Message structure:

Frame	Value	Description
Command	215	Command: FC_HEARTBEATREQUESTSTATUS
Method	0 or 1	Method : See table

Table: Methods

Method	Value	Description.
POLL_OLD	0	The server will reply with EV_HEARTBEATSTATUS_OLD. Deprecated
POLL	1	The server will reply with EV_HEARTBEATSTATUS. Use this object from protocolversion v2.04

EV_HEARTBEATSTATUS _OLD[70] (DEPRECATED)

General:

The server sends this message as reply to the request message.

Message structure:

Frame	Value	Description
Command	70	Command: EV_HEARTBEATSTATUS_OLD
Method	0	Method : Poll Old

EV_HEARTBEATSTATUS[72]

General:

The server sends this message as reply to the request message.

Message structure:

Frame	Value	Description
Command	72	Command: EV_HEARTBEATSTATUS
Method	1	Method : Poll

Home automation network management and status

Access Level

Home automation network management functions can only be executed with an additional access code.

There are four different access levels.
- No authority (default).
- User Access level.
- Installer Access level
- Engineer Access level.

In User access level mode the user can adjust simple settings. In Installer access level mode advanced network management functions can be executed. In Engineer access level mode the installer and user access passwords can be restored to default.

The access password format is four digits long. And is always send as four ASCII codes.
Default User password = '1111'.
Default Installer password = '2222'.

FC_ACCESSLEVELSET [222]

General:

A client uses this message to manage the Access level passwords.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	222	Command: FC_ACCESSLEVELSET
Method	Method	Method : See table
Data 0…N	0-255	Data f(Method)

Table: Methods

Method	Value	Description.
ATTR_CODE	0	Check the access level for a password. Data 0-3 holds the 4 digits of the password. This message doesn't adjust the current access level of the server it only verifies the access code can be used in the set function. A client user interface can use this to display a code access dialog an use the result of this message to build a new dialog depending the returned access level.
OBJECT_	1	Change a password.

CHANGECODE		Data 0-3 holds the old password, Data 4-7 holds the new password.
OBJECT_ RESET	2	Reset the password. Data 0-3 holds the password. When the password (Data0-3) is the installer password only the user password will be reset to defaults. When the password (Data0-3) is the engineer password both the User and the Installer password will be reset to defaults.

Example:

Situation	Frame.
Check the access level of password '1111'	[222,0,49,49,49,49]
Check the access level of password '2222'	[222,0,50,50,50,50]
Check the access level of password '1234'	[222,0,49,50,51,52]
Change the password from '1111' to '1234'	[222,1,49,49,49,49,49,50,51,52]
Reset the user password to '1111' using the default installer '2222' password.	[222,2,50,50,50,50]

EV_ACCESSLEVELINFO [75]

General:

The server sends this message as a reply to the access level set functions.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	75	Command EV_ACCESSLEVELINFO.
Method	Method	See table Methods
Data 0…N	0-255	Additional data.

Table: Methods

Method	Value	Description.
ATTR_CODE	0	The server responds with an access level. Data 0 = Access Level. (See table). Data 1-4= Code the client wanted to verify.
OBJECT_ CHANGECODE	1	The server responds with an access level.

		Data 0 = Access Level. (See Table - No Authority = Failed to change password.) Data 1-4= The new code.
OBJECT_ RESET	2	The server responds with status if the access codes are reset to default. Data 0 = 0 (=Failed) or 1 (=success).

Table: Access levels.

Access level	value
No Authority	0
Access level User	1
Access level Installer	2
Access level Engineer	3

Example:

Situation	Frame.
Password '1111' has access level User	[75,0,1,49,49,49,49]
Password '2222' has access level Installer	[75,0,2,50,50,50,50]
Password '1234' has access level No Authority	[75,0,0,49,50,51,52]
A new password '1234' for installer mode.	[75,1,2,49,50,51,52]
The password have been reset.	[75,2,1]

Node reset.

The server node can be reset via the TCP/IP protocol.

When a node in the CAN network is reset a message is broadcasted to all the clients. When the node holds the factory address the client application can ask the user to execute a plug and play.

FC_NODERESETSET [155]

General:

A client uses this message to reset the server node.
Attention: When this command is executed all clients will be disconnected.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	155	Command: FC_NODERESETSET
Method	Method	Method : See table
DATA0	48-57	Digit 1: User/Installer/Engineer password.
DATA1	48-57	Digit 2: User/Installer/Engineer password.
DATA2	48-57	Digit 3: User/Installer/Engineer password.
DATA3	48-57	Digit 4: User/Installer/Engineer password.

Table: Methods

Method	Value	Description.
OBJECT_RESET	0	The client uses this message to reset the server. An user/Installer or Engineer access code must be used.
OBJECT_RESET_FACTORY	1	The client uses this message to reset the server to factory default settings. All settings will be lost ! An Installer or Engineer access code must be used. The TFT application always cancels all running application tasks before executing this command. The client can do this via FC_APPLICATIONTASKSET + OBJECT_CANCEL.

Example:

Situation	Frame.
Reset the server node – Access code = default user code ‘1111’	[155,0,49,49,49,49]
Factory Reset the server node – Access code = default installer code ‘2222’	[155,1,50,50,50,50]

EV_NODERESETSET [18]

General:

The server sends this message when a node in the CAN-network has been reset. These messages are only send when the server is the master in the home automation network.

Attention: When the server node is reset there will be no EV_NODERESETSET send because all connections will be disconnected.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	18	Command: EV_NODERESETSET
Method	Method	Method : See table
DATA0	3-254	Logical node address

Table: Methods

Method	Value	Description.
OBJECT_ RESET	0	A node has been reset. Data 0: Holds the logical node address.
OBJECT_ RESET_ FACTORY	1	A node has been reset to the factory settings. (Or a new node has been added to the installation.) Data 0: Logical address = 254.
OBJECT_ RESET_ WATCHDOG	2	A node has been reset caused by a watchdog reset. Data 0: Holds the logical node address. This reset can happen when a hardware/software defect occurred.

Example:

Situation	Frame.
A node with logical address 0x0A has been reset.	[18,0,10]
A new node has been added to the installation.	[18,1,254]
A node with logical address 0x10 has been reset caused by a watchdog reset.	[18,2,16]

Home server node management.

Home automation CAN-network management is done via the master node. Only one node in the network can be assigned master.

Depending the software and protocol version different functions are available. The protocol and software version can be requested via these commands.
Attention: The commands to request this info are supported from protocol v2.04. (Also the possibility to set the tcp/ip server as master node is from this version.)

Not all hardware versions support to be set master node. Via these commands the client must verify that the server can be set master.

Before a client sets the server as the master node it must first verify the server holds the factory 0xFE address.

The client (application) must verify that all preconditions are set before configuring the server as the master node:

- Home automation network ok.
- Hardware compatibility: Can the tcp/ip server be configured as master node.
- Is there no other node In the network the master.
- Does the server hold the factory 0xFE logical address.
- …

To set the server as the master node an installer/Engineer access code must be used.

FC_NODEMANAGEMENTSET [219]

General:

The client must use this message to set the server as the master node in the CAN-network.
When the server is already the master node or when the hardware of the server is not compatible the server will always respond with a negative status to this command.

When the access code is wrong the server replies with a EV_MESSAGEERROR.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	219	Command FC_NODEMANAGEMENT
Method	0	OBJECT_SET_MASTER
DATA0	48-57	Digit 1: Installer/Engineer password.
DATA1	48-57	Digit 2: Installer/Engineer password.
DATA2	48-57	Digit 3: Installer/Engineer password.
DATA3	48-57	Digit 4: Installer/Engineer password.

Example:

Situation	Frame.
Set the server node as master – Default Installer password '2222'.	[219,1,50,50,50,50]

FC_REQUESTNODEMANAGEMENT [220]

The client must use this message to request info from the server node.

Message structure:

Frame	Value	Description
Command	220	Command FC_REQUESTNODEMANAGEMENT
Method	Method	See table Methods

Table methods:

Method	Value	Description
ATTR_MASTER_ SUPPORTED	0	Request if the server node has the feature to be set as master node.
ATTR_MASTER	4	Request if the server node is the master in the CAN network.
ATTR_NODEINFO	5	Request additional info from the server. Use this message to verify the server has a factory 0xFE address .
ATTR_VERSION_ SOFTWARE	6	Request the software version of the home server node.
ATTR_VERSION_ PROTOCOL	7	Request the protocol version of the home server node.

Example:

Situation	Frame.
Request if the server node supports the feature to be set as master	[220,0]
Request if the server node is the master.	[220,4]
Request additional node info.	[220,5]
Request Software version.	[220,6]
Request Protocol version	[220,7]

EV_NODEMANAGEMENTINFO [74]

General:

The server sends this message as a reply to the node management set and request messages.

Message structure:

Frame	Value	Description

Command	74	Command EV_NODEMANAGEMENTINFO
Method	Method	See table Methods
Data 0…N	0-255	Additional data.

Table methods:

Method	Value	Description
ATTR_MASTER_ SUPPORTED	0	The server responds to a ATTR_MASTER_SUPPORTED request message. Data 0: - 0=The server cannot be set master - 1=The server can be set master
OBJECT_SET_ MASTER_OK	2	The server responds to a set message. The server node has been changed from slave to the master in the home automation network.
OBJECT_SET_ MASTER_ERROR	3	The server responds to a set message. The server node has not been changed from slave to the master in the home automation network. Possible error: The server node was already the master. Possible error: There is already a master node in the home automation network.
ATTR_MASTER	4	The server response to a ATTR_MASTER Request message. Data 0 : - 0 = No master node. - 1 = Master node.
ATTR_NODEINFO	5	The server response to an ATTR_NODEINFO request message. Data 0-N = See table node info
ATTR_VERSION_ SOFTWARE	6	The server responds with the software version. Data 0: Major version Data 1: Minor version
ATTR_VERSION_ PROTOCOL	7	The server responds with the protocol version. Data 0: Major version Data 1: Minor version

Table node info:

Data Byte	Value	Description
Data 0	255	Don't care.
Data 1	3 - 252	Node logical address
Data 2	Ph address	H-MSB physical node address
Data 3	Ph address	H-LSB physical node address
Data 4	Ph address	L-MSB physical node address
Data 5	Ph address	L-LSB physical node address
Data 6	Length Name	Number of characters in Node name.
Data 7 … N	Name	Name data
Data N	0 - 254	Nr of Units.
Data N+1	4	Node Type = Gateway node.
Data N+2	Flags	Node flags: See table (advanced)

Table node flags:

Flags	Bitmask	Description
Master Controller	3	0: Normal node. 1: Master node.
System type	4-5	00 : Basic System. 10: Extended System. (only for gateway and LCD/TFT node.)
Audio System type	6	0: Basic Audio system. 1: Extended audio system. (only used by master node.)

Example:

Situation	Frame.
Reply master info – Master feature is supported.	[74,1,1]
Reply master info – Not the Master node	[74,4,0]
Reply master info – the Master node	[74,4,1]
Reply node info: Master Server Node 0xfc. Extended Mode.	[74,1,255,252,0,15,16,255,6,109,97,115,116,101,114,32,4,217]
Reply software version: Major = 101 (0x65) minor = 64 (0x40)	[74,6,101,64]
Reply protocol version: Major=2 (0x02) Minor=4 (0x04)	[74,7,2,4]

Application tasks.

An application tasks can only start when the server node is configured as the master in the home automation network.

A new task can only start when the application is in IDLE state.
When a task cannot be started and the server doesn't respond with a message error. This is probably because there is already a task running. The client connection must request the current running task.

When the client wants to start a new task with higher priority than the current running task the current running task must be cancelled and the connection must wait to receive state info = IDLE.

Example: When the server was reset and a node doesn't respond (maybe because the node has been removed from the installation) the UPDATE DATABASE task can take a long time to complete. The connection must first cancel this task before it can start the PNP task.

To execute an application task the set command must hold the installer or engineer access code.

FC_APPLICATIONTASKSET [212]

General:

The client must use this message to start or cancel an application task.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	212	Command FC_APPLICATIONTASKSET
Method	Method	See table Methods
DATA0	48-57	Digit 1: Installer/Engineer password.
DATA1	48-57	Digit 2: Installer/Engineer password.
DATA2	48-57	Digit 3: Installer/Engineer password.
DATA3	48-57	Digit 4: Installer/Engineer password.

Table methods:

Method	Value	Description
BROWSE	0	Start Browse for nodes task. The node database is NOT automatically erased by the server.
BROWSE_ CLEAR	1	Start Browse for nodes task. The node database is erased by the server.
PNP	2	Start Plug and play task. The node database is automatically erased by the server.
UPDATE_DBASE	3	Start Update database task
RESET_ALL	4	Reset all nodes. Only the nodes that are currently in the database.
FACTORY_ RESET_ALL	5	Reset all nodes to factory settings. Only the nodes that are currently in the database.

CANCEL	7	Cancel current task

Example:

Situation	Frame.
Start Browse for nodes – Default installer password = ‘2222’	[212,0,50,50,50,50]
Start Update dbase – Default installer password = ‘2222’	[212,2,50,50,50,50]
Cancel current task – Default installer password = ‘2222’	[212,4,50,50,50,50]

FC_REQUESTAPPLICATIONTASKSTATUS [213]

General:

The client must use this message to request if there are tasks running

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	213	Command FC_APPLICATIONTASKSET
Method	6	ATTR_STATE

Example:

Situation	Frame.
Request current state	[213,6]

EV_APPLICATIONTASKSTATUS [66]

General:

The server sends this message as response to the FC_APPLICATIONTASKSET message or spontaneous when the state of an application task changes.
If a connection started a task then only this connection will receive extended task info strings.

Message structure:

Frame	Value	Description
Command	66	Command EV_APPLICATIONTASKSTATUS
Method	Method	See table Methods
Data 0	0 - 255	Data

Data N	0 - 255	Data

Table methods:

Method	Value	Description
BROWSE	0	Send as response to FC_APPLICATIONTASKSET + BROWSE Browse for nodes task Status Data 0: 1= Started, 0=Not Started
BROWSE_ CLEAR	1	Send as response to FC_APPLICATIONTASKSET + BROWSE_CLEAR Browse for nodes task Status Data 0: 1= Started, 0=Not Started
PNP	2	Send as response to FC_APPLICATIONTASKSET + PNP PNP task Status Data 0: 1= Started, 0=Not Started
UPDATE_DBASE	3	Send as response to FC_APPLICATIONTASKSET + UPDATE_DBASE. Update database task Status Data 0: 1= Started, 0=Not Started
RESET_ALL	4	Send as response to FC_APPLICATIONTASKSET + RESET_ALL. Reset all task Status Data 0: 1= Started, 0=Not Started
FACTORY_ RESET_ALL	5	Send as response to FC_APPLICATIONTASKSET + FACTORY_ RESET_ALL. Reset all task Status Data 0: 1= Started, 0=Not Started
STATE	6	This is broadcasted to all connections when the task state changes. State info Data 0 : See table States
CANCEL	7	Send as response to FC_APPLICATIONTASKSET + CANCEL Cancel current task. Data 0: 1 = Cancelled, 0 = not cancelled.
INFO	8	Extended task Info. This info is only broadcasted to the client connection that started the task. Data 0 holds the length of the string message. Data 1-N holds formatted sting data.

Table states:

State	Value	Description
IDLE	0	No task running
BUSY_PNP	1	PNP task is running
BUSY_BROWSE	2	Browse for nodes task is running.
BUSY_UPDATEDBASE	3	Update database task is running
BUSY_GENERAL	4	General task is Busy: Example Reset All task.

Example:

Situation	Frame.
Status Task Browse Started	[66,0,1]
Status Task Browse failed to start	[66,0,0]

Application task Changed: Current task = PNP	[66,3,1]
Application task info – Length = 35 Data = "Browse for physical nodes again... "	[66,8,35,66,114,111,119,115,101,32,102,111,114,32,112,104,121,115,105,99,97,108,32,110,111,100,101,115,32,97,103,97,105,110,46,46,46,32]
Message Error - Homeserver is not the master node	[17,18]
Message Error – This client is not allowed to start/cancel a task.	[17,140]
Message Error – Wrong access code is used. Task will not be executed.	[17,141]

Node database management and status

Node database status:

The server holds a database of all nodes and units of the home automation network. Normal operation is only possible when the node database is ready.

Before the client can control the home automation network it must first verify the database state. When the database state changes the server will send the EV_NODEDATABASEINFO message.

When the server is the master in the CAN-network it is responsible to build the node database via browse for nodes algoritme. When the server is a slave in the CAN-network it's node database is updated automatically by the master node.

FC_NODEDATABASESET [221]

General:

The client must use this message to erase the servers node database.

The TFT application always cancels all running application tasks before executing this command. The client can do this via FC_APPLICATIONTASKSET + OBJECT_CANCEL.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	221	Command: FC_NODEDATABASESET
Method	6	Method OBJECT_ERASE
DATA0	48-57	Digit 1: Installer/Engineer password.
DATA1	48-57	Digit 2: Installer/Engineer password.
DATA2	48-57	Digit 3: Installer/Engineer password.
DATA3	48-57	Digit 4: Installer/Engineer password.

Example:

Situation	Frame.
Erase the server node database – Install password = '2222'	[221,6,50,50,50,50]

FC_NODEDATABASEREQUESTSTATUS [209]

General:

The client must use this message to request the state of the node database. Normally this message will be send after the login procedure has succeeded.

Message structure:

Frame	Value	Description
Command	209	Command: FC_NODEDATABASEREQUESTSTATUS
Method	5	Method ATTR_STATE

EV_NODEDATABASEINFO [64]

General:

The server sends this command as reply to the request message and also spontaneous when the state has changed. At any time the client must interpret this message.

Message structure:

Frame	Value	Description
Command	64	Command: EV_NODEDATABASEINFO
Method	5	Method ATTR_STATE
Data 0	Status	See table status

Table status:

Status	Value	Description.
EMPTY	0	The database is empty. There is no control possible
BUSY	1	The application is busy building the database. There is no control possible.
READY	2	The database is ready. Normal operation.

Examples:

Situation	Frame.
Database is empty	[64,5,0]
Database is busy	[64,5,1]
Database is ready	[64,5,2]

Iterating nodes and units: (Advanced)

To build a dynamic GUI all nodes and unit info can be requested.

The procedure holds the following steps:

1) Request number of nodes in the database.
2) Request info for all the nodes in the database.
3) Request info of all unit of individual nodes.

Step 1: Request the number of nodes.

FC_NODEDATABASEREQUESTSTATUS [209]

General:

The client must use this message to request the number of nodes in the database.

Message structure:

Frame	Value	Description
Command	209	Command: FC_NODEDATABASEREQUESTSTATUS
Method	0	Method ATTR_INFO

Example:

[209,0]

EV_NODEDATABASEINFO [64]

General:

The server sends this command as reply to the request message.

Frame	Value	Description
Command	64	Command: EV_NODEDATABASEINFO
Method	0	Method ATTR_INFO
Data 0	0 - 250	Number of nodes in the database.

Example:

Situation	Frame.
19 nodes in database	[64,0,19]

Step 2 : Request node info

FC_NODEDATABASEREQUESTSTATUS [209]

General:

The client must use this message to request the node info from the database. The index of the requested node must be in range of total number of nodes. (See previous section).

Message structure:

Frame	Value	Description
Command	209	Command: FC_NODEDATABASEREQUESTSTATUS
Method	1	Method ATTR_NODEINFO
Data0	0 - 250	Index of the node in the database.

Example:

Situation	Frame.
Request info first node	[209,1,0]

EV_NODEDATABASEINFO [64]

General:

The server sends this command as reply to the request message.

Frame	Value	Description
Command	64	Command: EV_NODEDATABASEINFO
Method	1	Method ATTR_NODEINFO
Data 0	0 - 250	Index of requested node
Data 1	3 - 252	Node logical address
Data 2	Ph address	H-MSB physical node address
Data 3	Ph address	H-LSB physical node address
Data 4	Ph address	L-MSB physical node address
Data 5	Ph address	L-LSB physical node address
Data 6	Length Name	Number of characters in Node name.
Data 7 ... N	Name	Name data
Data N	0 - 254	Nr of Units.
Data N+1	Type	Node Type: See table
Data N+2	Flags	Node flags: See table (advanced)

Table node types:

Node type	Value	Description
Standard	1	Standard node type.
Gateway	4	Gateway node, conversion from protocols.
Modem	8	Modem node only for configuration.

LCD, TFT	32	GUI node.

Table node flags:

Flags	Bitmask	Description
Master Controller	3	0: Normal node. 1: Master node.
System type	4-5	00 : Basic System. 10: Extended System. (only for gateway and LCD/TFT node.)
Audio System type	6	0: Basic Audio system. 1: Extended audio system. (only used by master node.)

Example:

Situation	Frame.
info first node, Node address = 15 Physical address = 123478 Name=A123478 Nr of Units = 32. Node type = Gateway System Type = Extended.	[64,1,0,15,0,18,52,120,7,65,49,50,51,52,55,56,32,4,129]
Error message : Node index 19 is invalid	[17,130,19]

Step 3: Request unit info

FC_NODEDATABASEREQUESTSTATUS [209]

General:

The client must use this message to request the info from a unit. The index of the requested unit must be in range of total number of units that the node holds.

Message structure:

Frame	Value	Description
Command	209	Command: FC_NODEDATABASEREQUESTSTATUS
Method	2	Method ATTR_UNITINFO
Data0	0 - 252	Logical address of the node.
Data 1	0 - 254	Index of the unit.

Example:

Situation	Frame.
Request info of first unit of node logical address 04	[209,2,4,0]

EV_NODEDATABASEINFO [64]

General:

The server sends this command as reply to the request message.

Frame	Value	Description
Command	64	Command: EV_NODEDATABASEINFO
Method	2	Method ATTR_UNITINFO
Data 0	3 - 252	Logical address of the requested node.
Data 1	0 - 254	Index of the requested unit.
Data 2	3 – 252	Logical node address
Data 3	0 – 254	Logical unit address
Data 4	Length Name	Length of the unit name.
Data 5 … N	Name	Name Data
Data N	Type	Unit type: See Table next section.
Data N+1	Flags	Unit flags: Ignore for the moment.

Example:

Situation	Frame.
Reply unit info: Logical address = 4,0 Unit type = UNIT_TYPE_SENS Name = ‘Sens1’	[64,2,4,0,4,0,5,115,101,110,115,49,4,1]

Error message: Node address 64 not found.	[17,129,64]
Error message: Node address 4 has no unit index 7.	[17,132,4,7]

Time and date.

The master node in the home automation network holds a real time clock. When the time changes this information is broadcasted to the slaves in the network.
A new time and date can only be set when the TCP/IP server is the master node.

FC_TIMEDATESET [170]

General:

The client must use this message to adjust the time and date.
When the server is not the master node it will return a message error.
When the real time clock is adjusted the server will respond with a confirmation message.

To adjust the time and date an user/installer or engineer access code is needed.

Message structure:

Frame	Value	Description
Command	170	Command FC_TIMEDATE
Method	0	Method ATTR_VALUE
Data0	48-57	Digit 1: User/Installer/Engineer password.
Data1	48-57	Digit 2: User/Installer/Engineer password.
Data2	48-57	Digit 3: User/Installer/Engineer password.
Data3	48-57	Digit 4: User/Installer/Engineer password.
Data4	0 - 23	Hour
Data5	0 - 59	Minutes
Data6	0 - 59	Seconds
Data7	0	Don't care
Data8	1 - 31	Day
Data9	1 - 12	Month
Data10	0 - 99	Century
Data11	0 - 99	Year

Example:

Situation	Frame.
15u38, 31 March 2011-Default user Access code '1111'	[170,0,49,49,49,49,15,38,0,4,31,3,21,11]

FC_REQUESTTIMEDATE [216]

General:

The client must use this message to request time and date info.

Message structure:

Frame	Value	Description
Command	216	Command FC_REQUESTTIMEDATE
Methods	0-3	Method ATTR_VALUE

Method	Value	Description.
ATTR_ VALUE	0	The client must use this message to request the current time and date. When the server is just start up this information can be wrong because it's possible that the server has not received correct time info.
ATTR_ SNTP_ ENABLED	3	The client must use this message to request if SNTP time synchronization is enabled. Only implemented for the nodes that support SNTP.

Example:

Situation	Frame.
Request time and date	[216,0]
Request SNTP enabled	[216,3]

EV_TIMEDATESTATUS [71]

General:

This message is send by the server as reply to a request message or spontaneous when the time or date (ATTR_VALUE) has changed.

Note:

The spontaneous event is not supported in the LITE protocol. In this version the client must explicit send a request message.

Message structure:

Frame	Value	Description
Command	71	Command EV_TIMEDATESTATUS
Method	0	Method ATTR_VALUE
Data0	0 - 23	Hour
Data1	0 - 59	Minutes
Data2	0 - 59	Seconds
Data3	1 - 7	Day of week: Monday =1, Sunday = 7
Data4	1 - 31	Day
Data5	1 - 12	Month

Data6	0 - 99	Century
Data7	0 - 99	Year

Message structure: Answer when a client sets Time/Date:

Frame	Value	Description
Command	71	Command EV_TIMEDATESTATUS
Method	1 or 2	1 = OBJECT_SET_OK 2 = OBJECT_SET_ERROR

Message structure: Answer when a client requests if SNTP is enabled.

Frame	Value	Description
Command	71	Command EV_TIMEDATESTATUS
Method	3	ATTR_SNTP_ENABLED
Data	0-1	0=disabled, 1=enabled

Example:

Situation	Frame.
15u38, Thursday 31 March 2011	[71,0,15,38,0,4,31,3,21,11]
Response to FC_TIMEDATESET Time and date is set	[71,1]
Response to FC_TIMEDATESET Time and date was not set	[71,2]

Week scheduling.

The master node in the home automation network holds a scheduling regime. The activation of temperature and time events depend on this scheduling regime.
Four different regimes can be set. When the regime changes this information is broadcasted to the slaves in the network.

FC_SCHEDULESET [217]

General:

The client must use this message to change week schedule regime.

Message structure:

Frame	Value	Description
Command	217	Command FC_SCHEDULESET
Method	0	Method ATTR_WEEKSCHEDULE
Data0	0 - 3	Regime (See Table.)

Example:

Situation	Frame.
Set to holiday mode	[217,0,3]

FC_REQUESTSCHEDULE [218]

General:

The client must use this message to request the current schedule regime.

Message structure:

Frame	Value	Description
Command	218	Command FC_REQUESTSCHEDULE
Method	0	Method ATTR_WEEKSCHEDULE

Example:

Situation	Frame.
Request regime	[218,0]

EV_SCHEDULESTATUS [73]

General:

This message is send by the server as reply to a request message or spontaneous when the schedule regime has changed.

Note:

The spontaneous event is not supported in the LITE protocol. In this version the client must explicit send a request message.

Message structure:

Frame	Value	Description
Command	73	Command EV_SCHEDULESTATUS
Method	0	Method ATTR_VALUE
Data0	0 – 3	Regime (See Table).

Example:

Situation	Frame.
Regime Holiday	[73,0,3]

Table Schedule regimes.

Status	Value
WEEK_0	0
WEEK_1	1
WEEK_2	2
HOLIDAY	3

Videophone. (Advanced).

The videophone mechanism is only possible in an Extended system and must be configured via the multimedia configuration. The server must be configured as indoor device otherwise no actions can be triggered nor will events be received.

The client must act like an indoor device and must handle correct upon videophone events.
When a call is made at an outdoor device all indoor devices are notified. The videophone mechanism sends a CALL event to all indoor devices. Only one indoor device can accept/discard the call. When an indoor device accepts/discards the call all indoor devices are notified. Only the device that accepted the call has control.

When a client from the server accepts the call the server becomes the active indoor device. The client that accepted the call becomes the active client. All other clients get notified a peer connection has accepted the call.

FC_VIDEOPHONESET [223]

General:

The client must use this message to trigger videophone actions.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	223	Command FC_VIDEOPHONESET
Method	0-255	Method See Table
Data	0-255	Optional data byte.

Table: Methods

Method	Value	Description.
HANGUP	2	Hang up the videophone call. This message only works when this indoor device is the active indoor device.
ACCEPT	3	Accept the videophone call. This is only possible when no other indoor device is active.
OPEN_DOOR	4	Open a door. Data : see table
DISCARD	5	Discard the videophone call. This is only possible when no other indoor device is active.

Table: Open door actions.

Method	Value	Description.
PULS	0	Short press on the open door button.

TOGGLE_ON	1	Long press on the open door button. Attention: After sending this command always send a TOGGLE_OFF command.
TOGGLE_OFF	2	Long press (Releasing) on the open door button. Attention: First send a TOGGLE_ON command.

Example:

Situation	Frame.
Accept the call	[223,3]
Hang up the call	[223,2]
Discard the call	[223,5]
Open the door - puls	[223,4,0]

EV_VIDEOPHONESTATUS [76]

General:

The server sends this message spontaneous when a videophone event occurs.

Note:

Not supported in LITE protocol.

Message structure:

Frame	Value	Description
Command	76	Command EV_VIDEOPHONESTATUS
Method	0-255	Method See Table
Data 0	252	Node Address, always 252
Data 1	255	Unit Address, always 255

Table: Methods

Method	Value	Description.
CALL	1	Call from an outdoor device. All indoor devices can accept/discard the call. The Client user interface can show a pop-up dialog that a call was made and show buttons: Accept, Discard and open door.
CALL_END	7	When the server node was the active indoor device this event is received when the call ends. This event is send to all clients. This event can also be send as acknowledge to a CALL_DISCARD method or because of a timeout. The client user interface can destroy the pop-up dialog.
CALL_START	8	When the server node accepted the call it becomes the active indoor device. This event is only send to the client connection that accepted the call. All other clients receive the CALL_START_PEER event. The client user interface can change its user interface.

		Only a Hang up and open door buttons are needed.
CALL_BUSY	9	When another indoor device accepted the call this message is send. The client can destroy the pop-up.
CALL_DISCARD	10	When an indoor device has discarded the call this message is send. This message is also send when the call request times out because no indoor device has accepted/discarded the call. The client can destroy the pop-up dialog.
CALL_START_ PEER	11	When the server accepted the incoming call this event is send to all other client connections that have not answered the request. The client can destroy the pop-up dialog.

Example:

Situation	Frame.
A call has been made at an outdoor device.	[76,1,252,255]
The call has been accepted by another device than the server node.	[76,9,252,255]
Acknowledge to the client that the call has started. The server and this client is the active indoor device.	[76,8,252,255]
Notifies the client that the server is the active indoor device but another client has accepted the request.	[76,11,252,255]

Registermap.

Since protocol version V2.09.
The master node in the system holds a register map with 1024 addresses. Each address in this map can hold a 32bit integer value. When the value in the register map changes the clients are notified.

FC_REGISTERMAP [224]

General:

This message can be send by the client to write or read a register address.

When the client writes a value to a register address the message is forwarded to the master in the CAN-bus system. When the value changes the system is notified about this change.
When the client reads a value from a register address the value in the register map of the server is returned. The message is not forwarded to the master in the system.

The addresses range from 0 to 1023. When the message holds a wrong address value an error message is returned.

NOTE: **Clients should not use the write function for the same address within an interval of 2 sec.** When a value is written to the master this value is stored in a queue. The master checks this fifo queue periodically and broadcast a status change to the system. When a value is written for a register address which is still in the queue the value is updated and only the last value will be used for the system.

Message structure

Frame	Value	Description
Command	224	Command FC_REGISTERMAP
Method	Method	See table: Read or write.
Address	0 – 3	MSB-Address
Address	0 – 255	LSB-Address
Value	0 – 255	HMSB value (not for ATTR_VALUE_READ).
Value	0 – 255	HLSB value (not for ATTR_VALUE_READ).
Value	0 – 255	LMSB value (not for ATTR_VALUE_READ).
Value	0 – 255	LLSB value (not for ATTR_VALUE_READ).

Table Methods:

Method	Value	Description
ATTR_VALUE_WRITE	1	Write a value to a register address. The message is forwarded to the master in the system. The message holds the address + 32bit value.
ATTR_VALUE_READ	2	Read a value from a register address. The server returns the current value in his own register map. The message holds the address.

Examples:

Situation	Frame.
Address 0: Read value	[224,2,0,0]
Address 1023: Read value	[224,2,3,255]
Address 0: Write value 200	[224,1,0,1,0,0,0,200]
Address 0: Write maximum value 2147483647	[224,1,0,0,127,255,255,255]
Address 0: Write minimum value -2147483648	[224,1,0,0,128,0,0,0]
Address 1023: Write value 0	[224,1,3,255,0,0,0,0]
Address 1023: Write value 50	[224,1,3,255,0,0,0,50]

EV_REGISTERMAP [77]

General:

This message is returned to the client as response to a read command. This message is send spontaneous to all the client when the value in a register address changes.

The addresses range from 0 to 1023

Message structure

Frame	Value	Description
Command	77	Command EV_REGISTERMAP
Method	0	ATTR_VALUE
Address	0 – 3	MSB-Address
Address	0 – 255	LSB-Address
Value	0 – 255	HMSB value
Value	0 – 255	HLSB value
Value	0 – 255	LMSB value
Value	0 – 255	LLSB value

Examples:

Situation	Frame.
address 0: status value 100	[77,0,0,0,0,0,0,100]
Address 10: status value 100	[77,0,0,10,0,0,0,100]
Address 10: status value 1000	[77,0,0,10,0,0,3,232]
Address 10: status value 10000	[77,0,0,10,0,0,39,16]
Address 10: status value -100	[77,0,0,10,255,255,255,156]
Address 10: status value -200	[77,0,1,0,255,255,255,56]
Address 1023: status value 100	[77,0,3,255,0,0,0,100]
Address 1023: status value 100000	[77,0,3,255,0,1,134,160]

Error message – Wrong address range.

When a read or write command is send to the server with a wrong address the server replies with an error message. The addresses range from 0 to 1023

Message structure

Frame	Value	Description
Command	77	Command EV_REGISTERMAP
Method	128	EC_WRONG_ADDRESS
Address	0 – 255	MSB-Address: The wrong address value
Address	0 – 255	LSB-Address: The wrong address value

Home automation control, status and request messages.

General:

A home automation unit is addressed via a node and unit address. Every type of unit has its own message functions. The messages send to a unit must match the unit type.

Every unit type can be controlled via a set function.
At every moment the client can request the status of a unit via a Request function. (for example at startup.) The server will reply with a message which holds the actual status of the unit.
When the status of a unit changes, the server will spontaneous send a message which holds the actual status of the unit. This status message is different for every type of unit.

When a set or request message is send to a unit of the wrong type an error message will be replied. (see section Error messages.)

Packet fields:

Request Message:

[Command , method ,node id, unit id, unit type]

Command Message:

[Command , method ,node id, unit id, data, data N-1]

Status Message:

[Command , method ,node id, unit id, unit type, data, data N-1]

Messages:

Request unit status:

General:

This message is send by the client to request the actual status of a unit. The server will reply with a status message. The status message is different for every unit type.

When the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	209	Command: FC_NODEDATABASEREQUESTSTATUS
Method	3	Method ATTR_UNITSTATUS
Node id	3 – 252	Node logical address
Unit id	0 - 255	Unit Logical address
Data 0	Unit Type.	See table unit types.

Table unit types:

Unit Type	Value	Description
UNIT_TYPE_DIM	1	Dimmer unit
UNIT_TYPE_SWITCH	2	Switch/Relay unit.
UNIT_TYPE_CONTROL	3	Control input unit.
UNIT_TYPE_SENS	4	Temperature sensor unit
UNIT_TYPE_AUDIO_EXT	5	Extended audio matrix unit.
UNIT_TYPE_VIRTUAL	7	Virtual, mood unit.
UNIT_TYPE_DUOSWITCH	8	Duo switch motor unit
UNIT_TYPE_AUDIO_BASIC	10	Basic audio unit.
UNIT_TYPE_AVMATRIX	11	AV Matrix unit
UNIT_TYPE_IRTX	12	IRTX unit
UNIT_TYPE_VIDEOMUX	14	Video multiplexer unit

Examples:

Situation	Frame.
Dimmer Unit: 5,2	[209,3,5,2,1]
Sensor unit 10,2	[209,3,10,2,4]
Virtual unit 252,0	[209,3,252,0,7]

Switch unit:

A switch unit can be controlled on/off.

FC_UNITSWITCHSET [163]

General:

This message can be send by the client to control a relay, switch unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	163	Command: FC_UNITSWITCHSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address

Table: Methods

Method	Value	Description/
OFF	2	Switch the device off
ON	3	Switch the device on
FSCVONOFF	4	Toggle the device on/off

Examples:

Situation	Frame.
Device 3,0: Switch on	[163,3,3,0]
Device 3,0: Switch off	[163,2,3,0]
Device 3,0: Toggle on/off	[163,4,3,0]

EV_UNITSWITCHSTATUS [6]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	6	Command: EV_UNITSWITCHSTATUS
Method	0	Method: STATUS
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	2	UNIT_TYPE_SWITCH

Data 0	config	Reserved, ignore at the moment.
Data 1	Status	See table status.

Table status:

Status	Value	Description
OFF	0	The device is switched off
ON	1	The device is switched on
PIRTIMED	2	The device is switched on via PIR detection.

Examples:

Situation	Frame.
Device 3,0: Status on	[6,0,3,0,2,0,1]
Device 3,0: Status off	[6,0,3,0,2,0,0]
Device 3,0: Status PIR-timed	[6,0,3,0,2,0,2]

Dimmer unit:

A dimmer unit can be switch on/off and the dimmer value can be controlled.

FC_UNITDIMSET [162]

General:

This message can be send by the client to control a dimmer unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	162	Command: FC_UNITDIMSET
Method	Method	See table methods.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	Value (1-99)	Only for ATTR_VALUE.

Table methods:

Method	Value	
OFF	9	Switch the device off
ON	10	Switch the device on
FSCVONOFF	11	Toggle the device on/off
DIMUP	5	Dim the device up. (Attention: When dimming is done must send ON)
DIMSDOWN	6	Dim the device down. (Attention: When dimming is done must send ON).
DIMFS	7	Dim the de device up or down f(current state). (Attention: When dimming is done must send ON).
ATTR_VALUE	3	Switch device on with a dimmer value. (Data 0). Dimmer value 1-99. See Note !!!

Note:

When the light is switched off and the ATTR_VALUE method is send the status (EV_UNITDIMSTATUS) to this message will not hold the correct dimmer value.
Solution 1: When the dimmer is off before sending the ATTR_VALUE method first send an ON command.
Solution 2: After sending the ATTR_VALUE method send a FC_UNITDIMREQUESTSTATUS message to request the dimmer value.

Examples:

Situation	Frame.
Device 4,1: Switch on	[162,10,4,1]
Device 4,1: Switch on with value 75%	[162,3,4,1,75]
Device 4,1: Switch off	[162,9,4,1]

EV_UNITDIMSTATUS [5]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	5	Command: EV_UNITDIMSTATUS
Method	0	Always 0
Node id	3 – 250	Node logical address
Unit id	0 - 255	Unit Logical address
Unit type	1	UNIT_TYPE_DIM: Always 1 for a dimmer unit.
Data 0	State Ext Flags	See table: Table Extended state flags
Data 1	Status	See table: Table Status
Data 2	Dim value	Current dimmer value: 1-99

Table status:

Status	Value	Description
OFF	0	The device is switched off
ON	1	The device is switched on
PIRTIMED	2	The device is switched on via PIR detection.

Table Extended state flags:

Bit	Description
0	When bit is set: Feedback Error. DALI – Dimmer Unit: Indicates lamp failure or slave device not in system (=not responding).
1	Reserved
2	Reserved
3	Reserved
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Examples:

Situation	Frame.
Device 4,1 State = Off	[5,0,4,1,1,0,0,75]
Device 4,1 State = On, value = 75%	[5,0,4,1,1,0,1,75]

FC_UNITDIMREQUESTSTATUS [131]

General:

This message can be send by the client to request the dimmer value. The server will send this request message to the dimmer unit. The dimmer unit will respond and the server will send a EV_UNITDIMSTATUS message.
This is sometimes needed when a dimmer is switched on via the FC_UNITDIMSET – ATTR_VALUE command.

When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	131	Command: FC_UNITDIMREQUESTSTATUS
Method	Method	See table methods.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address

Table methods:

Method	Value	Description
ATTR_VALUE	3	Request current dimmer value.

Examples:

Situation	Frame.
Dimmer 7,0	[131,3,7,3]

Virtual and control unit:

A virtual or mood unit can hold scenarios that were recorded by the configuration software. When this unit is triggered by a puls or long event the recorded actions are executed. A virtual unit can also be used to help in binding configuration. It can be used as condition, visualize state, invoke binding actions,…
Examples: After 22h, Dark outside, …

A control unit is an input unit. This input can be physical connected to input controls. (PIR detection, door bell control, …)
This unit can be controlled by its physical input and also via software. This unit cannot hold scenarios but can be configured to invoke actions.

FC_UNITCONTROLSET [168]

General:

This message can be send by the client to control a control and virtual unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	168	Command: FC_UNITCONTROLSET
Method	Method	See table.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	0 or 1	Only for method LONG

Table Methods:

Method	Value	
PULS	2	Short puls event
LONG	3	Long toggle event: Data : - OFF = 0, - ON = 1

Examples:

Situation	Frame.
Unit 5,2 : puls event	[168,2,5,2]
Unit 5,2 : Long event + On	[168,3,5,2,1]
Unit 5,2 : Long event + Off	[168,3,5,2,0]

EV_UNITCONTROLSTATUS [4]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Attention: A control and virtual unit will only return a feedback status when this is configured via the binding configuration software. The feedback status can be used to display the status of a condition. (After 22h, Dark outside, …)

Message structure:

Frame	Value	Description
Command	4	Command EV_UNITCONTROLSTATUS.
Method	0	Always 0
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	3 or 7	UNIT_TYPE_CONTROL = 3, UNIT_TYPE_VIRTUAL = 7
Data 0	config	Reserved, ignore for the moment.
Data 1	Status	See table.

Table Feedback status:

Status	Value	Description
OFF	0	Feedback status = off
ON	1	Feedback status = on.

Examples:

Situation	Frame.
Control unit 5,2 : feedback status off	[4,0,5,2,3,0,0]
Virtual unit 252,2 : feedback status on	[4,0,252,2,7,0,1]

Motor unit:

A motor unit is a duo switch that can be triggered in two directions: up or down.
When the motor is running it will automatically stop after a time and go to idle state.
There is also a small time between switching directions.
This two timers are configured in the home automation network.

FC_UNITDUOSWITCHSET [182]

General:

This message can be send by the client to control a duo switch unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	182	Command: FC_UNITDUOSWITCHSET
Method	Method	See table methods.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address

Table methods:

Method	Value	Description
STOP	3	Stop the motor
UP	4	Start motor direction up
DOWN	5	Start motor direction down
FSUPDOWN	6	Start motor + toggle direction.

Examples:

Situation	Frame.
Unit 10,2 Start the motor direction up.	[182,4,10,2]
Unit 10,2 Stop the motor.	[182,3,10,2]

EV_UNITDUOSWITCHSTATUS[38]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	38	Command: EV_UNITDUOSWITCHSTATUS.
Method	0	Always 0
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	8	UNIT_TYPE_DUOSWITCH
Data 0	Config	Reserved, ignore for the moment
Data 1	Status	See table.

Table status:

Status	Value	Description
IDLE	0	Motor is stopped, don't know last direction. Possible after reset
IDLE_DOWN	1	Motor is stopped, last direction was down
IDLE_UP	2	Motor is stopped, last direction was up.
BUSY_DOWN	3	Motor is busy, direction is down
BUSY_UP	4	Motor is busy, direction is up.

Examples:

Situation	Frame.
Unit 10,2 Motor is at the moment busy up	[38,0,10,2,8,0,4]
Unit 10,2 Motor has stopped, last dir was up	[38,0,10,2,8,0,2]

Temperature sensor unit:

The sensor unit can be used to control heating and cooling. It has four presets that hold a separate setpoint value. At regular intervals the actual temperature is measured and compared with the setpoint value. The sensor unit can go to the states idle, heating and cooling.
The sensor unit can be switched on/off.

To be able to control a third party airco/heating system there have been properties added to the existing sensor unit. These properties can only be controlled when the sensor unit is part of a temperature gateway node. These properties are not implemented by our standard nodes.

General

Temperature values (setpoints and measured values) are send as 16 bit two's complement (16 bit short signed value) values.
Attention: **Negative values: bit 15 holds a sign bit !!**

Temperature value = (256*MSB + LSB) / 10

Examples:

Temperature	Unsigned value	MSB	LSB
15°C	150	0	150
22°C	220	0	220
25,5°C	255	0	255
26°C	260	1	4
30°C	300	1	44
33,5°C	335	1	79
-5,0°C	65468	255	206
-10,0°C	65436	255	156
-15,0°C	65386	255	106
-40°C	65136	254	112

Pseudo example code:

```
typedef short int16_t;

char szText[16];
const int16_t value=(MSB<<8)+(int16_t)LSB;
sprintf(szText,"Value=%i.%i°C",value/10,abs(value%10));
```

or

```
char szText[16];
int value =(MSB<<8)+LSB;
if (value & 0x8000) {
	value = (value^0xffff)+1;
	sprintf(szText,"Value=-%i.%i°C",abs(value/10),abs(value%10));
}
else{
	sprintf(szText,"Value=%i.%i°C",abs(value/10),abs(value%10));
}
```

FC_UNITSENSSET [136]

General:

This message can be send by the client to control a sensor unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure:

Frame	Value	Description
Command	136	Command: FC_UNITSENSSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit logical address
Data 0	Optional	See method table: used for SETPOINT, ATTR_ONOFF and ATTR_PRESET.
Data 1	Optional	See method table: used for SETPOINT.
Data 2	Optional	See method table: used for SETPOINT.

Table methods

Method	Value	Description
SETPOINT	1	Set preset setpoint: Data0: - Sun=0, - Half sun =1, - Moon=2, - Half moon = 3, Data1: - MSB setpoint data. Data 2: - LSB setpoint data
ATTR_ ONOFF	3	Set sensor on/off : Data0: - OFF = 0, - ON = 1
SETPOINT_ INC	5	Increase current preset setpoint
SETPOINT_ DEC	6	Decrease current preset setpoint
FSONOFF	9	Toggle sensor on/off
ATTR_ PRESET	13	Select preset: Data0: - Sun=0, - Half sun =1,

		- Moon=2, - Half moon = 3,
ATTR_ WORKINGMODE (**)	16	Data 0: Values 0 – 5 and 255. See Appendix 1: Working modes. More modes will be added to the protocol when needed
ATTR_ FANSPEED (**)	17	Data 0: Values 0 – 5 and 255 See Appendix 1: Speed values. More speed values will be added to the protocol when needed
ATTR_ SWINGANGLE (**)	18	Data 0: MSB swing angle Data 1: LSB swing angle The values should be between 0 – 360 See Appendix1 : Swing angle and Position Values.
ATTR_ SWINGMODE (**)	19	Data 0: - Swing mode off = 0, - Swing mode On = 1

(**): Attention:
These new methods are only implemented when the sensor unit is part of the temperature gateway. When these methods are send to a unit that doesn't implement these properties there will be no error message.

(**): Attention:
The functions depend on the HVAC system: See Appendix 1.

Examples:

Situation	Frame.
Sensor unit 5,1 preset moon and setpoint = 18,5°C.	[136,1,5,1,2,0,185]
Sensor unit 5,1 preset sun and setpoint = 22,5°C	[136,1,5,1,0,0,225]
Sensor unit 5,1 preset sun and setpoint = 28,5°C	[136,1,5,1,0,1,29]
Sensor unit 5,1 Off.	[136,2,5,1,0]
sensor unit 5,1 Increase setpoint	[136,5,5,1]
Sensor unit 5,1 Toggle on/off	[136,9,5,1]
Sensor unit 5,1 preset Half sun.	[136,13,5,1,1]
Sensor unit 3,1 Set working mode Auto	[136,16,3,1,0]
Sensor unit 3,1 Set working mode Fan	[136,16,3,1,4]
Sensor unit 3,1	[136,18,3,1,0,20]

Set swing angle 20°	

FC_UNITREQUESTSENSSTATUS [137]

General:

This message can be send by the client to request an extended sensor unit property. The server will forward this message to the sensor unit via the domotics control bus.
When the sensor unit implements the extended unit properties an extended unit status message will be replied when the server node receives the respond from the sensor unit.
When the sensor unit doesn't implement these properties no message will be replied. On the domotics control bus a message error will be replied.
Therefore it is recommended the client application knows which sensor units implement these properties before the request message is send.

Message structure

Frame	Value	Description
Command	137	Command: FC_UNITREQUESTSENSSTATUS
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit logical address

Table methods

Method	Value	Description
ATTR_ WORKINGMODE	16	Request current working mode
ATTR_ FANSPEED	17	Request current fan speed
ATTR_ SWINGANGLE	18	Request current swing angle
ATTR_ SWINGMODE	19	Request current swing mode.

Examples:

Situation	Frame.
Sensor unit 3,1 Request swing angle	[137,18,3,1]
Sensor unit 3,1 Request working mode	[137,16,3,1]

EV_UNITSENSSTATUS[7]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	7	Command EV_UNITSENSSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	4	UNIT_TYPE_SENS
Data 0	config	See table sensor types:
Data 1	Control state on/off	Current control state on/off: off = 0, on = 1
Data 2	State	Current state: idle = 0, Heating = 1, Cooling = 2
Data 3	Preset	Current preset: Sun =0, Half sun=1, Moon=2, Half moon =3
Data 4	Curr. measured value	MSB measured value
Data 5	Curr. measured value	LSB measured value
Data 6	Setp. value sun	MSB preset sun – setpoint
Data 7	Setp. Value sun	LSB preset sun – setpoint
Data 8	Setp. value half sun	MSB preset half sun – setpoint
Data 9	Setp. value half sun	LSB preset half sun – setpoint
Data 10	Setp. value moon	MSB preset moon – setpoint
Data 11	Setp. value moon	LSB preset moon – setpoint
Data 12	Setp. value half moon	MSB preset half moon – setpoint
Data 13	Setp. value half moon	LSB preset half moon – setpoint.

Message structure: (Extended)

This extended message status is only send by a sensor unit when one of the extended properties changes or is requested by the FC_UNITREQUESTSENSSTATUS request message.
It is not the answer to the general unit request message and it doesn't replace the standard status message. When one of the clients sends a FC_UNITREQUESTSENSSTATUS request message all the servers connections will receive a status message.

General rule: The client application must at least once request these parameters to have a correct user interface update.

Frame	Value	Description

Command	7	Command EV_UNITSENSSTATUS
Method	1	Method STATUS_EXTENDED
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	4	UNIT_TYPE_SENS
Data 0	config	See table sensor types:
Data 1	Control state on/off	Current control state on/off: off = 0, on = 1
Data 2	State	Current state: idle = 0, Heating = 1, Cooling = 2
Data 3	Preset	Current preset: Sun =0, Half sun=1, Moon=2, Half moon =3
Data 4	Curr. measured value	MSB measured value
Data 5	Curr. measured value	LSB measured value
Data 6	Setp. value sun	MSB preset sun – setpoint
Data 7	Setp. Value sun	LSB preset sun – setpoint
Data 8	Setp. value half sun	MSB preset half sun – setpoint
Data 9	Setp. value half sun	LSB preset half sun – setpoint
Data 10	Setp. value moon	MSB preset moon – setpoint
Data 11	Setp. value moon	LSB preset moon – setpoint
Data 12	Setp. value half moon	MSB preset half moon – setpoint
Data 13	Setp. value half moon	LSB preset half moon – setpoint.
Data 14	Measured offset val (**)	MSB measured offset value, default = 0
Data 15	Measured offset val (**)	LSB measured offset value, default = 0
Data 16	Swing angle (**)	MSB Swing angle value, default = 0 See Appendix1 : Angle RTD Net and Position Values.
Data 17	Swing angle (**)	LSB Swing angle value, default = 0 See Appendix1 : Angle and Position Values.
Data 18	Working mode (**)	Values 0 – 5 and 255. See Appendix 1: Working modes.
Data 19	Fan speed (**)	Values 0 – 5 and 255. See Appendix 1: Speed Values.
Data 20	Swing mode (**)	Off=0, on=1.

(**) Indicates the extended sensor properties.

Sensor types:

Type	value	Description
TEMPERATURE	0	Temperature control: Heating and cooling.
PH	1	Reserved - Not implemented
LUX	2	Reserved - Not implemented

AMPERE	3	Reserved - Not implemented

Example:

Situation	Frame.
Sensor unit 252,0 State On, Heating, current preset = SUN, Measured value = 20,0°C setpoint Sun = 21°C, setpoint half sun = 22°C, setpoint moon = 16°C setpoint half moon = 20°C	[7,0,252,0,4,0,1,1,0,0,200,0,210,0,220,0,160,0,200]
Extended message: Sensor unit 3,1 State on, Cooling Current preset = SUN Measured value = 29,5° Setpoint Sun = 20° Setpoint half sun = 21° Setpoint moon = 17° Setpoint half moon = 28° Measured offset val = 0° Swing angle = 90° Working mode = Cooling Fan speed = slow Swing mode = off	[7,1,3,1,4,0,1,2,0,1,39,0,200,0,210,0,170,1,24,0,0,0,90,2,0,0]

EV_UNITSENSSTATUS[7] (Advanced)

General:

This message can be send by a 3rd party sensor to send the measured value to the sensor. The home automation system will use this measured value to control cooling/heating and display it in the user interfaces.

Message structure:

Frame	Value	Description
Command	7	Command EV_UNITSENSSTATUS
Method	10	Method: Measured value.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	4	UNIT_TYPE_SENS
Data 0	Measured value	MSB measured value
Data 1	Measured value	LSB measured value

Examples:

Situation	Frame.
Sensor unit 252,123 Measured value -10°C	[7,10,252,123,4,255,156]
Sensor unit 252,123 Measured value 15.5°C	[7,10,252,123,4,0,155]
Sensor unit 252,123 Measured value 22.0°C	[7,10,252,123,4,0,220]

Attention: Values are send as two complements 16 bit values.

Example: -10°C is send as 255,156 (FF9C)

Pseudo example code:

```
const int negativeValue = ((measured_value * 10)^0xFFFF) - 1;
const int MSB = (negativeValue & 0xFF00) >> 8;
const int LSB = (negativeValue & 0xFF);
```

Basic audio unit:

The Basic audio unit is used to control an audio unit. The unit represents an audio room that can have maximum one destination and maximum 8 sources.

FC_UNITAUDIOBASICSET [159]

General:

This message can be send by the client to control a Basic audio unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure

Frame	Value	Description
Command	159	Command FC_UNITAUDIOBASICSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	Optional	Used by SOURCE_SEL and SOURCE_FC.

Table: Methods:

Method	Value	Description
SOURCE_SEL	1	Source selection Data0 : Source 0 – 7 = 0 - 7
ON	4	Switch audio on with current source
OFF	5	Switch audio off
FSCSONOFF	7	Switch audio off or On with current source depending on the current state.
VOLUME_UP	8	Volume audio up. (Attention: Must send ON when volume set is done.)
VOLUME_DN	9	Volume audio down. (Attention: Must send ON when volume set is done.)
FSVOLUPDOWN	10	Set volume up or down depending on the current state. (The room must be on state.) When the room is switched on, volume up will be first executed, next volume down. (Attention: Must send ON when volume set is done.)
PRESS_UP	11	Source function: Press up
PRESS_DOWN	12	Source function: Press down
SOURCE_FC	14	Source function: Data 0: - function Play = 0, - function Pause = 1
SOURCE_ EXTRA_FC	23	Extra source function: (only for type BOSE_V35) Data 0: function: see table Extra source functions. Actions depend on the current source playing.

SOURCE_PLAY	32	source function play: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_STOP	33	source function stop: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_PAUSE	34	source function pause: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_NEXT	35	source function next: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_PREV	36	source function prev: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_FFWD	37	source function ffwd: (only for type BOSE_V35) Actions depend on the current source playing.
SOURCE_FRWD	38	source function frwd: (only for type BOSE_V35) Actions depend on the current source playing.

Examples:

Situation	Frame.
Audio unit 15,8 Select Source 3.	[159,1,15,8,3]
Audio unit 15,8. Switch audio with current source	[159,4,15,8]
Audio unit 15,8. Switch audio off.	[159,5,15,8]

Table: Extra Source functions

Method	Value	Description
Keypad 1	0	Keypad digit 1
Keypad 2	1	Keypad digit 2
Keypad 3	2	Keypad digit 3
Keypad 4	3	Keypad digit 4
Keypad 5	4	Keypad digit 5
Keypad 6	5	Keypad digit 6
Keypad 7	6	Keypad digit 7
Keypad 8	7	Keypad digit 8
Keypad 9	8	Keypad digit 9
Keypad 0	9	Keypad digit 0
Keypad up	10	Keypad up
Keypad down	11	Keypad down
Keypad left	12	Keypad left
Keypad right	13	Keypad right
Keypad enter	14	Keypad enter
Keypad dash	15	Keypad dash
Dvd menu	16	DVD menu
Guide	17	Guide
Info	18	Info
Exit	19	Exit menu

free	20	Not used - reserved
Up1	21	Red (Only Bose V535 and newer systems)
Down1	22	Green (Only Bose V535 and newer systems)
Up2	23	Yellow (Only Bose V535 and newer systems)
Down2	24	Blue (Only Bose V535 and newer systems)

EV_UNITAUDIOSTATUS[23]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	23	Command EV_UNITAUDIOSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id (**)	0 – 255	Unit Logical address
Unit type	5 or 10	UNIT_TYPE_AUDIO_BASIC 5 or UNIT_TYPE_AUDIO_RS232 10.
Data 0	Config	Audio system type: see table
Data 1	Source	Current source 0 -7, OFF = 255
Data 2	volume	Current volume: 0 – 100, unknown = 255
Data 2	current disc info	Current disc info. Disc: 0 – 8, Unknown = 255
Data 3	Current track info	Current track info: Track: 0 – 31 Unknown = 255
Data 4	Tuner Preset info	Current tuner preset. Unknown = 255
Data 5	Tuner frequency	MSB When unknown tuner frequency = 65536
Data 6	Tuner frequency	LSB When unknown tuner frequency = 65536

Attention: (**)
When the unit id value is 255 this means the status message is for all the units of this type for this node id. These status messages are used when the tuner frequency/preset or disc/track info is changed.

Table audio system type:

Type	Value	Unit type	Description
AUDIO_DT	0	5	Simple Duotecno audio system (Basic system). Maximum 8 sources.
BOSE_LF_50	1	10	Bose Lifestyle 50 system. 6 sources.
BOSE_LF_48	2	10	Bose Lifestyles systems LF18,LF28,LF38,LF48 and V30 8 sources
BOSE_LF_V35	3	10	Bose Lifestyle systems: V25,V35, T10 and T20. 8 sources.

Examples:

Situation	Frame.
Audio unit 15,8. Bose V35 system, Room Off	[23,0,15,8,10,3,255,25,255,255,255,255,255]
Audio unit 15,8 Bose V35 system, Room On, Current playing Source 3, volume =25	[23,0,15,8,10,3,3,25,255,255,255,255,255]

Attention:

The fields volume, disc, track, tuner preset and frequency depend on the audio system type. When a the system doesn't have the information for this fields the unknown values are returned.
See table:

Examples:
- Type BOSE_LF_48 will always return unknown values for Volume, disc and track info.
- After a reset the Type BOSE_LF_48 system will always send unknown values for tuner preset and frequency. As soon as the user changes tuner presets via the Bose remote the system will return true values.
- Type BOSE_LF_V35 will always return unknown values for disc and track info

System	**Volume**	**Disc Info**	**Track Info**	**Tuner Preset**	**Frequency**
AUDIO_DT	255	255	255	255	65536
BOSE_LF50	0…99 or 255	0…8 (*)	0…31 (*)	0 … 254.	FM – band
BOSE_LF48	255	255	255	0 … 8 (*). or 255.	0 = AM, 1 = FM, or 65536
BOSE_V35	0…99 or 255	255	255	0 … 24 (*)	AM – band FM – band or 65536

(*) : Must add +1 to display a valid value.

Frequency conversion Bose LF50:

sprintf(szText, "FM %d.%02d P--",Freq/256,Freq%256)

Frequency conversion Bose LF48:

No info available for tuner frequency: AM = 0, FM = 1

Frequency conversion Bose V35 – AM Band

```
const int MIN_FREQUENCY_AM = 55;
const int MAX_FREQUENCY_AM = 1710;
sprintf(szText,"AM %d", Freq)
```

Frequency conversion Bose V35 – FM Band

```
const unsigned long MIN_FREQUENCY_FM = 76000ul;
const unsigned long MAX_FREQUENCY_FM = 107900ul;

unsigned long ulTemp = ((unsigned long)(( Freq /256))*1000) +( Freq %256);
if (ulTemp >= MIN_FREQUENCY_FM && ulTemp<= MAX_FREQUENCY_FM)
{
  sprintf(szText,"FM %d.%02d",specs.m_usTunerFreq/256,specs.m_usTunerFreq%256);
}
```

Extended audio matrix unit

The extended audio matrix unit is used to control an audio unit. The unit represents an audio room that can have maximum four destinations and maximum 8 sources.

FC_UNITAUDIOEXTSET [208]

General:

This message can be send by the client to control a matrix audio unit.
When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure

Frame	Value	Description
Command	208	Command FC_UNITAUDIOEXTSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	Destination NR	0 … 3
Data 1	Optional	Used by SOURCE_SEL and SOURCE_FC methods.

Table Methods:

Method	Value	Description
SOURCE_SEL	1	Select a source. Data 0: Destination Nr Data 1: Source Nr 0…7
ON	4	Switch dest on with current source Data 0: Destination Nr
OFF	5	Switch dest off Data 0: Destination Nr
FSCSONOFF	7	Switch audio off or On with current source depending on the current state. Data 0: Destination Nr
VOLUME_UP	8	Volume up. (Must send VOLUME_STOP when volume set is done.) Data 0: Destination Nr
VOLUME_DN	9	Volume down. (Must send VOLUME_STOP when volume set is done.) Data 0: Destination Nr
VOLUME_STOP	10	Volume stop. Data 0: Destination Nr
FSVOLUPDOWN	11	Set volume up or down depending on the current state. (The room must be on state.) When the room is switched on, volume up will be first executed, next volume down. (Attention: Must send ON when volume set is done.) Data 0: Destination Nr

VOLUME	12	Send an absolute volume. (Volume range depends on AV matrix device.) Data 0: Destination Nr Data 1: Volume
PLAY	15	Source function Data 0: Destination Nr
STOP	16	Source function Data 0: Destination Nr
PAUSE	17	Source function Data 0: Destination Nr
FF	18	Source function Data 0: Destination Nr
RR	19	Source function Data 0: Destination Nr
FFWD	20	Source function Data 0: Destination Nr
FRWD	21	Source function Data 0: Destination Nr
SOURCE_FC	24	Advanced Source functions Data 0: Destination Nr Data 1: Source function - see table extra source functions.

Table extra source functions:

Key	Value	Description
KP_1	0	Keypad button 1
KP_2	1	Keypad button 2
KP_3	2	Keypad button 3
KP_4	3	Keypad button 4
KP_5	4	Keypad button 5
KP_6	5	Keypad button 6
KP_7	6	Keypad button 7
KP_8	7	Keypad button 8
KP_9	8	Keypad button 9
KP_0	9	Keypad button 0
KP_UP	10	Keypad menu up
KP_DOWN	11	Keypad menu down
KP_LEFT	12	Keypad menu left
KP_RIGHT	13	Keypad menu right
KP_ENTER	14	Keypad menu enter
KP_10	15	Keypad button -/--
KP_FREE1	16	Keypad free function 1
KP_FREE2	17	Keypad free function 2
KP_FREE3	18	Keypad free function 3
KP_FREE4	19	Keypad free function 4
KP_FREE5	20	Keypad free function 5
UP1	21	Keypad extra up function
DOWN1	22	Keypad extra down function

UP2	23	Keypad extra up function
DOWN2	24	Keypad extra down function

Examples:

Situation	Frame.
Audio unit 252,64 Audio destination 2, Select Source 3.	[208,1,252,64,2,3]
Audio unit 252,64 Audio destination 2, Set volume up	[208,8,25,64,2]
Audio unit 252,64. Audio destination 2, current source Keypad digit 5	[208,24,252,64,2,4]

EV_UNITAUDIOEXTSTATUS[70]

General:

This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	70	Command EV_UNITAUDIOSTATUS.
Method	0	Method STATUS
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	5	UNIT_TYPE_AUDIO = 5
Data 0	Config	Reserved
Data 1	Current selected dest	Current selected destination. Unknown = 255
Data 2	Dest 0 – source	Dest 0 : Current source 0 – 7, OFF = 255
Data 3	Dest 1 – source	Dest 1 : Current source 0 – 7, OFF = 255
Data 4	Dest 2 – source	Dest 2 : Current source 0 – 7, OFF = 255
Data 5	Dest 3 – source	Dest 3 : Current source 0 – 7, OFF = 255
Data 6	Dest 0 – volume	Dest 0 : Current volume
Data 7	Dest 1 – volume	Dest 1 : Current volume
Data 8	Dest 2 – volume	Dest 2 : Current volume
Data 9	Dest 3 – volume	Dest 3 : Current volume

Examples:

Situation	Frame.
Audio unit 252,64	[70,0,252,64,5,0,2,255,2,2,255,25,50,0,35]

Current selected destination 2. Status dest 0 = off, Status dest 1 = source 2, volume 50 Status dest 2 = source 2, volume 0 Status dest 3 = off	

IRTX unit. (Advanced)

The IRTX unit is used to transmit IR codes.

FC_UNITIRTXSET [173]

General:

This message can be send by the client to control a IRTX unit.

When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Normally the home automation system holds intelligence when to send an IRTX code. Via this protocol one can bypass this intelligence and send additional IRTX codes.

Message structure

Frame	Value	Description
Command	173	Command FC_UNITIRTXSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	MSB – IRTX code	Not for method STOP
Data 1	LSB – IRTX code	Not for method STOP

Table Methods:

Method	Value	Description
START	1	Start sending an IRTX code. This can be used to send IRTX codes for volume control. - Data 0: MSB IRTX code - Data 1: LSB IRTX code.
STOP	2	Stop sending an IRTX code.
PULS	3	Send a IRTX code only once. - Data 0: MSB IRTX code - Data 1: LSB IRTX code.

Calculation of IRTX code:
IRTX code = (256*MSB) + LSB.

Examples:

IRTX Code	MSB	LSB
10	0	10
255	0	255
256	1	0
300	1	44

Examples:

Situation	Frame.
IRTX unit: 32,53 Send IR-Code 10.	[173,3,32,53,0,10]
IRTX unit: 11,14 Send IR-Code 255	[173,3,11,14,0,255]
IRTX unit: 11,14 Send IR-Code 256	[173,3,11,14,1,0]
IRTX unit: 11,14 Send IR-Code 300	[173,3,11,14,1,44]

EV_UNITDEFAULTSTATUS [48]

General:

This is a default message and holds no important parameters.
This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	48	Command EV_UNITDEFAULTSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	12	UNIT_TYPE_IRTX = 12

IRRX unit (Advanced).

The IRRX unit receives IR and request the master node to check if the received command is mapped.
The master node checks in the database if the code is mapped and executes an action or sends a result back to the IRRX unit.
When the IRRX unit receives a result from the master node it can trigger actions via the binding mechanism.

Note: There are no messages to control the IRRX unit.

FC_CHECKIRRXCODE [192]

General:

This message is send when the homeserver is the master node and an IRRX unit received an IR code.
This event is useful for configuration purposes or maybe to trigger external events of a third party system.
Attention: The homeserver must be the master node to send this command.

Note:

Not supported in LITE protocol.

Message structure

Frame	Value	Description
Command	192	Command FC_CHECKIRRXCODE
Method	Method	See table
Node id	3 – 252	Node logical address of IRRX receiver unit.
Unit id	0 – 255	Unit Logical address of IRRX receiver unit.
Data 0	MSB – IRTX code	Received code.
Data 1	LSB – IRTX code	Received code.

Table Methods:

Method	Value	Description
TOGGLE_OFF	0	Long press: button released
TOGGLE_ON	1	Long Press: button pressed
PULS	2	Short press.

EV_UNITDEFAULTSTATUS [48]

General:

This is a default message and holds no important parameters.
This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	48	Command EV_UNITDEFAULTSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	13	UNIT_TYPE_IRRX = 13

Video multiplexer unit. (Advanced)

The video multiplexer unit is used to control a video multiplexer system. The system can be simple control via relay contacts or can be an extended system. The extended system can have controls for digital recording and playback.

FC_UNITVIDEOMUXSET [193]

General:

This message can be send by the client to control a video multiplexer unit.

When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Message structure

Frame	Value	Description
Command	193	Command FC_UNITVIDEOMUXSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	F(Method)	Data varies function of method
Data 1	F(Method)	Data varies function of method

Table Methods:

Method	Value	Description
CONNECT	02	Connect input to output: - Data 0 = output. - Data 1 = input
DISCONNECT	03	Disconnect input from output: - Data 0 = output. - Data 1 = input
DISCONNECT_ALL	04	Disconnect all Cameras.
KEYPRESS	10	Only for Digital video multiplexer systems. Simulate a key press. - Data 0 : See table key press functions.
KEYPRESS_CAMERA	11	Only for Digital video multiplexer systems. Simulate a key press: - Data 0 : Camera 0 – 15
DVMR	12	Only for Digital video multiplexer systems. Digital video recorder/playback. - Data 0 : See table DVMR functions.

Table key press functions

Name	Value

MULTISCREEN	0
ENTER	1
MENU	2
TILT_UP	3
TILT_DOWN	4
PAN_LEFT	5
PAN_RIGHT	6
ZOOM	7
SEQUENCE	8
FREEZE	9
ALARM_ACK	10

Table DVMR functions

Name	Value
LIVE_MODE	0
REVCORD_MODE	1
PLAY_MODE	2
STOP	3
PLAY_FORWARD	4
PLAY_REVERSE	5
FAST FORWARD	6
FAST REVERSE	7

EV_UNITDEFAULTSTATUS [48]

General:

This is a default message and holds no important parameters.
This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	48	Command EV_UNITDEFAULTSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	14	UNIT_TYPE_VIDEOMUX = 14

AV Matrix unit. (Advanced)

The AV Matrix unit is used to control an audio video matrix system. These a/v matrix systems are mostly connected via a RS232 interface to a Duotecno interface node. The Duotecno extended audio unit holds intelligence to control this unit.

Since firmware version v65.50 (protocol version 2.06) the server can hold an AV Matrix unit. Via this A/V matrix unit it is possible to control a 3rd party Audio Video matrix via the TCP/IP open protocol. The home automation audio room has to be controlled by the audio commands FC_UNITAUDIOEXTSET and EV_UNITAUDIOEXTSTATUS. To control the 3rd party a/v equipment the client only has to implement the EV_UNITMACROCOMMAND events and translate these to send commands to the connected AV matrix equipment.
If needed extra status information can be send to the home automation by sending the EV_UNITAVMATRIXSTATUS command.

FC_UNITAVMATRIXSET [202]

General:

This message can be send by the client to control a AV Matrix unit.

When a wrong method is send or the unit cannot be found in the node database the server replies with a message error. (See section error messages)

Normally the home automation system holds intelligence when to send an AV Matrix command. Via this protocol one can bypass this intelligence and send additional commands. This should only be done in special circumstances.

Message structure

Frame	Value	Description
Command	202	Command FC_UNITAVMATRIXSET
Method	Method	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Data 0	Output	F(Method)
Data 1	Input / volume	F(Method)
Data 2	State	F(Method)

Table Methods:

Method	Value	Description
ATTR_ CONNECTION	1	Connect or disconnect input to output: - Data 0 = output. - Data 1 = input - Data 2 = state 0 or 1
CONNECT	02	Connect input to output: - Data 0 = output. - Data 1 = input

DISCONNECT	03	Disconnect input from output: - Data 0 = output. - Data 1 = input
VOLUME	04	Set matrix output volume - Data 0 = output. - Data 1 = volume (range depends on type).
BASS	05	Set matrix output bass volume - Data 0 = output. - Data 1 = volume (range depends on type).
TREBLE	06	Set matrix output treble volume - Data 0 = output. - Data 1 = volume (range depends on type).
VOLUME_INC	07	inc matrix output volume (step = +1) - Data 0 = output. - Data 1 = volume
VOLUME_DEC	08	Dec matrix output volume (step = -1) - Data 0 = output. - Data 1 = volume
DISCONNECT_ ALL	09	Disconnect all inputs from outputs.
VOLUME_ UP	10	Start matrix output volume up - Data 0 = output.
VOLUME_ DOWN	11	Start matrix output volume down - Data 0 = output.
VOLUME_ STOP	12	Stop matrix output volume control - Data 0 = output.
VOLUME_ MUTE_ TOGGLE	17	Toggle matrix volume mute. - Data 0 = output.
MACRO	18	Send a macro command to the AV matrix system. (Not supported for all types of matrix) - Data 0 = output - Data 1 = MSB – macro code. - Data 2 = LSB – macro code.

Examples:

Situation	Frame.
AV Matrix unit: 18,0. Set matrix output 8 volume 20	[202,04,18,0,8,20]

EV_UNITDEFAULTSTATUS [48]

General:

This is a default message and holds no important parameters.
This message is send by the server as reply to a status request message or spontaneous when a parameter changes.

Message structure:

Frame	Value	Description
Command	48	Command EV_UNITDEFAULTSTATUS
Method	0	Method STATUS.
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	11	UNIT_TYPE_AVMATRIX= 11

EV_UNITAVMATRIXSTATUS [54]

General:

When the audio room is controlled by the FC_UNITAUDIOEXTSET commands this is not needed.

This command can only be send when the homeserver holds an AVMatrix unit.
The client can use this command to send extra status information to the A/V matrix unit. Via this unit the status of the audio room in the home automation system will be updated. This is possible needed when a 3rd party control system controls the AV matrix equipment.

Note that the server will not broadcast EV_UNITMACROCOMMAND messages to its clients when the changes are triggered by this status change message.

Message structure:

Frame	Value	Description
Command	54	Command EV_UNITAVMATRIXSTATUS
Method	1 or 4	See table
Node id	3 – 252	Node logical address
Unit id	0 – 255	Unit Logical address
Unit type	11	UNIT_TYPE_AVMATRIX= 11
Data 0	0 - 15	Output number
Data 1	F(Method)	Data varies function of method
Data 2	F(Method)	Data varies function of method

Table Methods:

Method	Value	Description
CONNECTION	1	(Dis) Connection - Data 0: Output number: 0 – 15 - Data 1: Input number: 0-7, 255 is unknown. - Data 2: 0=disconnect, 1=connect
VOLUME	4	Volume value - Data 0: Output number: 0 – 15 - Data 1: Value 0 – 50

Examples:

Situation	Frame.
AV Matrix unit: 32,51	[54,01,32,51,11,01,02,01]

Status update Connection : Output 1, Input 2	
AV Matrix unit: 32,51 Status update disconnect: Output 0, input 1	[54,01,32,51,11,00,01,00]
AV Matrix unit: 32,51 Status update Connection : Output 15, Input 2	[54,01,32,51,11,15,02,01]
AV Matrix unit: 32,51 Status volume: Output 0, Volume 40	[54,04,32,51,11,00,40]

Error messages

Error messages will be send by the server when wrong parameters are send by the client or when the application is not ready to process the client request.

EV_MESSAGEERROR [17]

General:

This message will be send by the server.

Message structure

Frame	Value	Description
Command	17	Command EV_MESSAGEERROR
Method	Method	See table
Data 0	F(Method)	Data varies function of method
Data 1	F(Method)	Data varies function of method

Table Methods:

Method	Value	Description
WRONG_ OBJECT_ METHOD	11	Wrong object method received. - Data 0 = Node address - Data 1 = unit address - Data 2 = Object member.
WRONG_ MESSAGECODE	12	Wrong Message Code received. - Data 0 = Message Code. Since protocol version v2.04
ONLY_FOR_ MASTER	18	This function can only be executed when this node is the master. Since protocol version v2.04
NODEDATABASE_ NOT_READY	128	The node database is not ready. Command or request cannot be processed by the server.
NODEADDRESS_ NOT_ FOUND	129	Node with this address could not be found in the database. - Data 0 = node address
NODEINDEX_ NOT_ FOUND	130	Wrong node index. - Data 0 = node index. Possible caused because there are less nodes in the database then the requested index.
UNITADDRESS_ NOT_ FOUND	131	Unit with this address could not be found in the database. - Data 0 = node address. - Data 1 = unit address.
UNITINDEX_ NOT_ FOUND	132	Wrong unit index. - Data 0 = node address - Data 1 = unit index.

		Possible caused because the node holds less units then the requested index.
UNIT_ WRONG_ TYPE	133	Unit is of different type. - Data 0 = node address - Data 1 = unit address.
OPERATION_ NOT_ ALLOWED	140	The requested operation is not allowed. This error message can be send to a client that wants to cancel an application task algoritme but is not allowed to do this. For example when another connection started the operation…
WRONG_ ACCESSCODE	141	The requested operation is not allowed because a wrong access (installer) code is used.
NOT_ IMPLEMENTED	142	The requested operation is not implemented in this software version. Possible because the node only supports the LITE protocol version.

Macro command messages. (Advanced).

When the home server node holds units, these units can trigger macro commands.
These macro commands can be used to control a third-party system.

EV_UNITMACROCOMMAND [69]

General:

This message will be send by the server when the state of the unit changes. For the moment not all unit types are supported.

Note:

Not supported in LITE protocol.

Message structure

Frame	Value	Description
Command	69	Command EV_UNITMACROCOMMAND
Method	Method	0
Data 0	0 – 252	Logical node address
Data 1	0 - 254	Logical unit address
Data 2	Event	F(unit type) : Event – See table.
Data 3	State	F(unit type) : State - See table
Data 4	Code	F(unit type) : MSB – code.
Data 5	Code	F(unit type) : LSB – code.

Table: UNIT_TYPE_DIM

Name	Event	State	Code
OFF	6	0	--
ON	6	1	0 - 99
PIR timed	7	1	0 - 99
PIR Timed off	7	0	--
Dim up	8	2	0 – 99 (*)
Dim down	8	3	0 – 99 (*)
Dim stop	6	1	0 - 99

(*) Since protocol version 2.07: The dimmer values have been added in the events Dim down & Dim Up.

Table: UNIT_TYPE_SWITCH

Name	Event	State	Code
OFF	4	0	--
ON	4	1	--
PIR timed	5	1	--
PIR timed off	5	0	--

Table: UNIT_TYPE_DUOSWITCH

Name	Event	State	Code
Stop	32	0	--
Up	32	1	--
Down	32	2	--

Since protocol version 2.07.

Table: UNIT_TYPE_IRTX

Name	Event	State	Code
PULS	1	1	0 - 65536

Table: UNIT_TYPE_CONTROL en UNIT_TYPE_LCD_VIRTUAL

Name	Event	State	Code
SHORT_OFF	3	0	--
SHORT_ON	3	1	--
LONG_OFF	2	0	--
LONG_ON	2	1	--

Table: UNIT_TYPE_SENS

Name	Event	State	Code
CONTROL_OFF	9	0	--
CONTROL_ON	9	1	--
STATE_OFF	10	0	--
STATE_HEATING	10	1	--
STATE_COOLING	10	2	--
SETPOINT_SUN	11	0	Temp value * 10
SETPOINT_HALFSUN	11	1	Temp value * 10
SETPOINT_MOON	11	2	Temp value * 10
SETPOINT_HALFMOON	11	3	Temp value * 10
WORKINGMODE_AUTO	12	0	--
WORKINGMODE_HEATING	12	1	--
WORKINGMODE_COOLING	12	2	--
WORKINGMODE_DRY	12	3	--
WORKINGMODE_FAN	12	4	--
FANSPEED_SPEED_1 (SLOW)	13	0	--
FANSPEED_SPEED_2 (HIGH)	13	1	--
FANSPEED_SPEED_N	13	N-1 (**)	--
FANSPEED_AUTO	13	255	--
SWINGANGLE	14	1	Degrees 0 – 360
SWINGMODE_OFF	15	0	--
SWINGMODE_ON	15	1	--

(**) N: For the moment the fan speed is limited to 5.

Table: UNIT_TYPE_AVMATRIX

Name	Event	State	Code
POWER_ON	20	Output nr: 0-15	--
CONNECT	21	Output nr: 0-15	Input nr: 0-7
DISCONNECT	22	Output nr: 0-15	Input nr: 0-7 or 255 (unknown) (If needed use this for POWER OFF)
VOLUME_UP	23	Output nr: 0-15	Value: 0-50 Send at periodic (120msec) intervals
VOLUME_DOWN	24	Output nr: 0-15	Value: 0-50 Send at periodic (120msec) intervals
VOLUME_STOP	25	Output nr: 0-15	Value: 0-50
VOLUME_VALUE	26	Output nr: 0-15	Value: 0-50
VOLUME_MUTE	27	Output nr: 0-15	0 = mute off, 1=mute on
MACRO	28	Output nr: 0-15	Macro code: 0-65536
DISCONNECT_ALL	29	--	--

Examples:

Situation	Frame.
Virtual unit 15,10: SHORT_ON	[69,0,15,10,3,1,0,0]
IRTX unit 15,11: IRTX code 10	[69,0,15,11,1,1,0,10]
Switch unit 15,12: RESET	[69,0,15,12,255,1,0,0]

Appendix 1: Extended sensor unit functions:

Working modes:

	Value	Daikin RTDNET	Daikin KKRP01	Daikin RTDNET2	Daikin RTDRA	Mitsubishi Procon A1M	Daikin VRVVKM	Samsung Intesis	Samsung Intesis ERV	Mitsubishi Procon A1M 2
WORKING_MODE_AUTO	0	Auto	Auto	Auto	Auto	Auto	-	Auto	Auto	Auto
WORKING_MODE_HEATING	1	Heating	Heating	Heating	Heating	Heating	-	Heating	HRV	Heating
WORKING_MODE_COOLING	2	Cooling	Cooling	Cooling	Cooling	Cooling	-	Cooling	Cleaning	Cooling
WORKING_MODE_DRY	3	Dry	Dry	Dry	Dry	Dry	-	Dry	Sleep	Dry
WORKING_MODE_FAN	4	Fan	Fan	Fan	Fan	Fan	-	Fan	Normal	Fan
WORKING_MODE_OFF	0xFF	-	-	-	-	-	-	-	-	-

Fan speed:

	Value	Daikin RTDNET	Daikin KKRP01	Daikin RTDNET2	Daikin RTDRA	Mitsubishi Procon A1M	Daikin VRVVKM	Samsung Intesis	Samsung Intesis ERV	Mitsubishi Procon A1M 2
FAN_SPEED_1	0	Slow	Speed 1	-	Speed 1	Quiet	Slow	Low	-	Quiet
FAN_SPEED_2	1	High	Speed 2	-	Speed 2	Weak	High	Middle	Middle	Weak
FAN_SPEED_3	2	-	Speed 3	-	Speed 3	Strong	-	High	High	Strong
FAN_SPEED_4	3	-	Speed 4	-	Speed 4	Strong ++	-	-	Turbo	Strong ++
FAN_SPEED_5	4	-	Speed 5	-	Speed 5	-	-	-	-	-
FAN_SPEED_AUTO	0xFF	-	Auto	-	Auto	Auto	-	Auto	-	Auto

Swing angle & position:

	Value	Daikin RTDNET	Daikin KKRP01	Daikin RTDNET2	Daikin RTDRA	Mitsubishi Procon A1M	Daikin VRVVKM	Samsung Intesis	Samsung Intesis ERV	Mitsubishi Procon A1M 2
Value 0 / ANGLE_RTDNET_0	0	0°	-	0°	-	Pos Auto	0°	Stop	-	-
Value 1	1					Pos 1		Up/Down		
Value 2	2					Pos 2		Left/Right		
Value 3	3					Pos 3		Both		
Value 4	4					Pos 4				
Value 5	5					Pos 5				
ANGLE_RTDNET_20	20	20°	-	20°	-	-	20°	-	-	-
ANGLE_RTDNET_45	45	45°	-	45°	-	-	45°	-	-	-
ANGLE_RTDNET_70	70	70°	-	70°	-	-	70°	-	-	-

ANGLE_RTDNET_90	90	90°	-	90°	-	-	90°	-	-	-

Swing mode on/off:

	Value	Daikin RTDNET	Daikin KKRP01	Daikin RTDNET2	Daikin RTDRA	Mitsubishi Procon A1M	Daikin VRVVKM	Samsung Intesis	Samsung Intesis ERV	Mitsubishi Procon A1M 2
SWINGMODE_OFF	0	-	Swingmode Off	-	Swingmode Off	-	-	-	-	-
SWINGMODE_ON	1	-	Swingmode On	-	Swingmode On	-	-	-	-	-